

BHARATH KUMAR KONDUR

GENERATIVE AI ENGINEER | LLM, RAG & AGENTIC SYSTEMS

+39 3891485034 | bharathkumarkondur@gmail.com | Rome, Italy | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI Engineer with 4+ years of experience across applied AI and machine learning, including building Generative AI applications, RAG systems, multimodal AI, computer vision and OCR solutions. Strong hands-on experience with Python, FastAPI, LangChain, LangGraph, vector databases, LLM evaluation, Docker and Kubernetes. Built production workflows combining structured retrieval, semantic search, tool calling and grounded LLM generation. Developed and deployed a production OCR model achieving ~95% recognition accuracy and built Thenfold, a live local-first GenAI/RAG application with hybrid retrieval, privacy-focused AI processing and secure multi-user architecture.

CORE SKILLS

Generative AI: Generative AI: RAG, Agentic AI, Tool Calling, Structured Outputs, Prompt Engineering, Context Engineering, LLM Evaluation, LLM-as-a-Judge.

LLM & Retrieval: LangChain, LangGraph, Langfuse, OpenRouter, Hugging Face, Ollama, Pinecone, MongoDB Atlas Vector Search, ChromaDB, Embeddings, Semantic Search, Hybrid Retrieval.

Backend & Data: Python, FastAPI, TypeScript, Next.js, REST APIs, WebSockets, PostgreSQL, Supabase, MongoDB, Redis, SQL

Cloud & MLOps: AWS (EKS, EC2, S3, Bedrock), Azure, Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Linux, Vercel, Prometheus, Grafana.

ML & computer Vision: PyTorch, TensorFlow, Scikit-learn, OCR, Object Detection, Multi-Object Tracking, Vision AI, ONNX Runtime, TensorRT, CUDA

PROFESSIONAL EXPERIENCE

Project Consulting Srl | Rome, Italy

AI & Computer Vision Engineer | *Generative AI / Applied AI Focus* | 11/2022 - 06/2026

- Built a production conversational AI assistant for a video-security and incident-intelligence platform, enabling natural-language access to operational events and internal knowledge.
- Designed Python/FastAPI microservices integrating LLM workflows with MongoDB, REST/WebSocket APIs, asynchronous processing and existing AI services.
- Implemented RAG pipelines combining structured MongoDB retrieval with semantic search using MongoDB Atlas Vector Search and ChromaDB; evaluated Pinecone for larger-scale retrieval.
- Built LangChain/LangGraph workflows for tool calling, structured-data retrieval, vector search, conversation state and grounded answer generation.
- Implemented Langfuse tracing/evaluation for latency, token usage, cost, tool-calling behaviour and source-grounded quality checks using LLM-as-a-Judge plus manual validation.
- Resolved GPU-memory pressure during concurrent local inference by reducing context length from ~128K to 16K tokens and introducing concurrency controls.
- Developed and deployed a custom vehicle licence-plate OCR model from scratch, achieving ~95% recognition accuracy, and operated AI workloads with Docker/Kubernetes, CI/CD and Prometheus/Grafana.

ENEA | Rome, Italy

Machine Learning Intern - Energy Systems & LCOE Estimation | 09/2021 - 07/2022

- Developed and evaluated machine-learning models for renewable-energy LCOE estimation and compared predictions using quantitative error metrics.

PROJECTS | 07/2026 - PRESENT

Thenfold - Local-First GenAI / RAG Memory Platform | *Live production application* | [Link](#)

- Designed and shipped a full-stack GenAI/RAG product using Next.js, TypeScript, Supabase/PostgreSQL, Pinecone, OpenRouter and Vercel, with separate local and cloud AI execution paths.
- Implemented MiniLM semantic embeddings, Pinecone retrieval, owner-scoped PostgreSQL keyword fallback and deterministic temporal filtering for first/latest, before/after and date-range memory questions.
- Built privacy and trust controls including AES-256-GCM local encryption, Supabase RLS, tenant-isolated vector namespaces, ZDR cloud routing and citation validation that rejects invalid source references.
- Validated authentication, cross-user isolation, retrieval, citations and AI workflows with 157 automated tests plus Playwright and live-production checks.

EDUCATION & CERTIFICATIONS

Sapienza University of Rome | Rome, Italy

MSc Energy Engineering | 07/2022 | Thesis: Improving Accuracy in LCOE Estimation using Machine Learning

JNTUH | Hyderabad, India

BTech Electrical & Electronics Engineering | 06/2016

Certifications: AWS Certified AI Practitioner (2026); Claude Code in Action - Anthropic (2026); DevOps Foundations: DevSecOps - LinkedIn (2025)

Languages: English (Fluent), Italian (B2), Hindi, Telugu